Www.Medicalstudyzone.com

ACID BASE BALANCE

$CO_2 + H_2O \xrightarrow[\text{Anhydrase}]{\text{Carbonic}} H_2CO_3 \text{ (Weak acid)} \longrightarrow H^+ \text{ (Acid)} + HCO_3^- \text{ (Base)}$

Maintainence of Acid Base Balance

00:01:00

pH :

Normal : 7.35 – 7.45 (> 7.8 or < 6.9 → Death).

Derangement of pH →
- Demineralization of proteinaceous enzymes (↓ activity).
- Alkalosis : ↑ CNS excitability
- Acidosis : ↓ CNS excitability

FIRST LINE DEFENSE :

Maintains pH without correcting cause of imbalance.

Extracellular buffer :

$CO_2 + H_2O \xrightarrow{CA} H_2CO_3 \longrightarrow H^+ + HCO_3^-$

Most important buffer : Tightly regulates CO_2 levels in the lung and H^+ in the kidney.

Intracellular buffer :

a) Proteins with histidine act as buffers.

Eg : Hb in RBC's (most plentiful intracellular buffer).

b) Phosphate : Urinary buffer.

Note : Ammonia → Urinary buffer.

SECOND LINE DEFENSE :

Maintains pH by altering pCO2 in lungs (Can correct upto 75%).

Alveolar ventilation $\alpha \frac{1}{pCO_2}$

Hyperventilation → ↑CO_2 washout → ↓pCO_2

Hypoventilation → ↑CO_2 → ↑pCO_2

gmail.com

Www.Medicalstudyzone.com

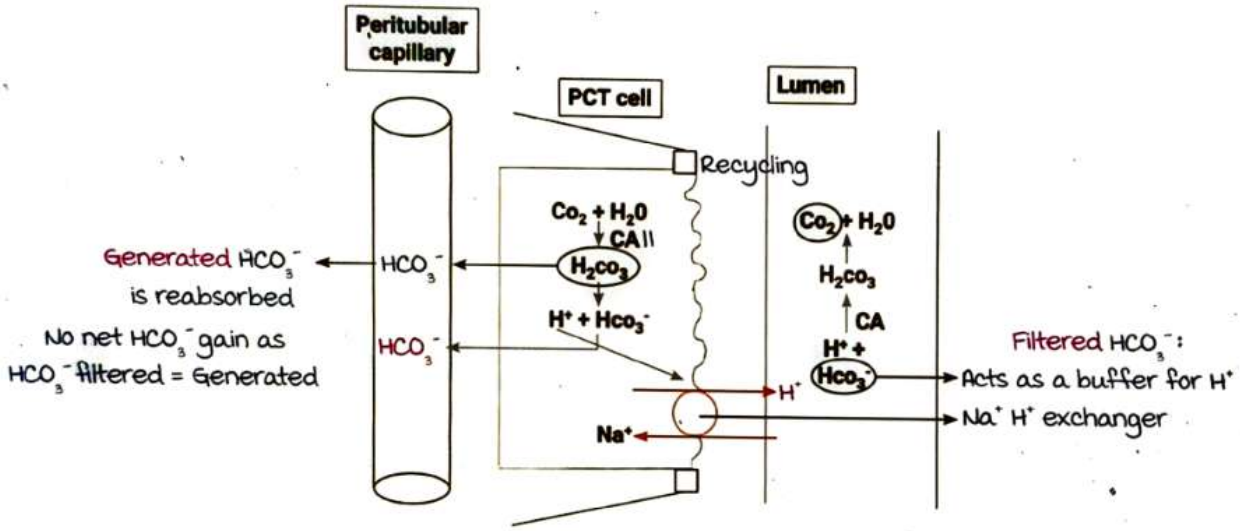
THIRD LINE DEFENSE :

----- Active space -----

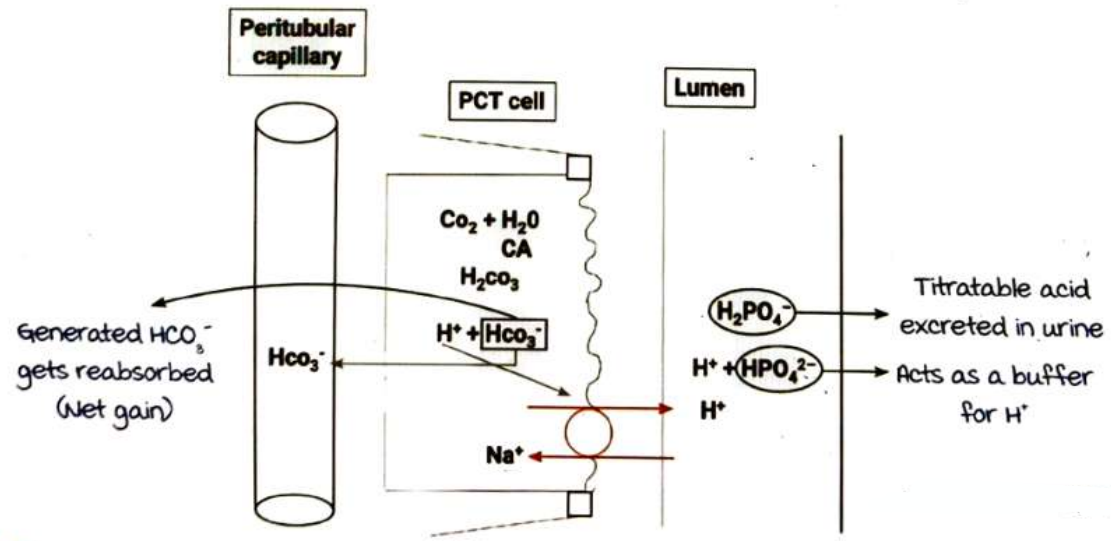
- Kidneys → Correct the cause → Normalize pH by :
  - Generation of HCO_3^- : Unique feature for Kidney.
  - H^+ excretion.
  - HCO_3^- excretion/reabsorption.

PCT :

1. Excretion of H^+ and reabsorption of HCO_3^- :



2. Generation of new HCO_3^- by using substitute buffers (Unique to Kidney).
   a. With HPO_4^{2-} as buffer.



Note :

Limiting pH :

pH at which PCT stops excretion of H^+ = 4.5 (Never seen in PCT due to strict buffering).

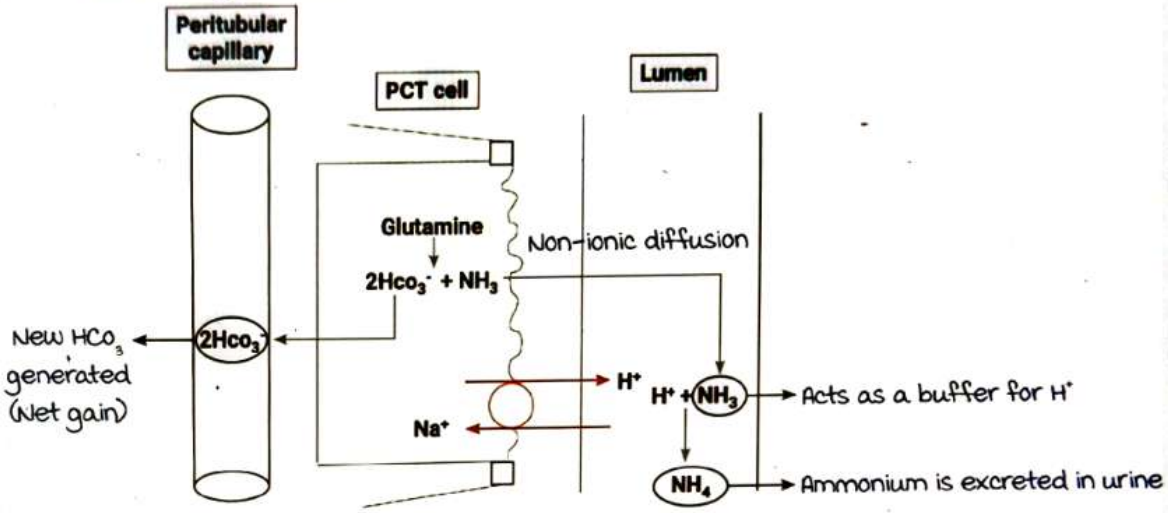
Seen in collecting duct, where urine is acidified.

Feedback

Www.Medicalstudyzone.com

----- Active space -----

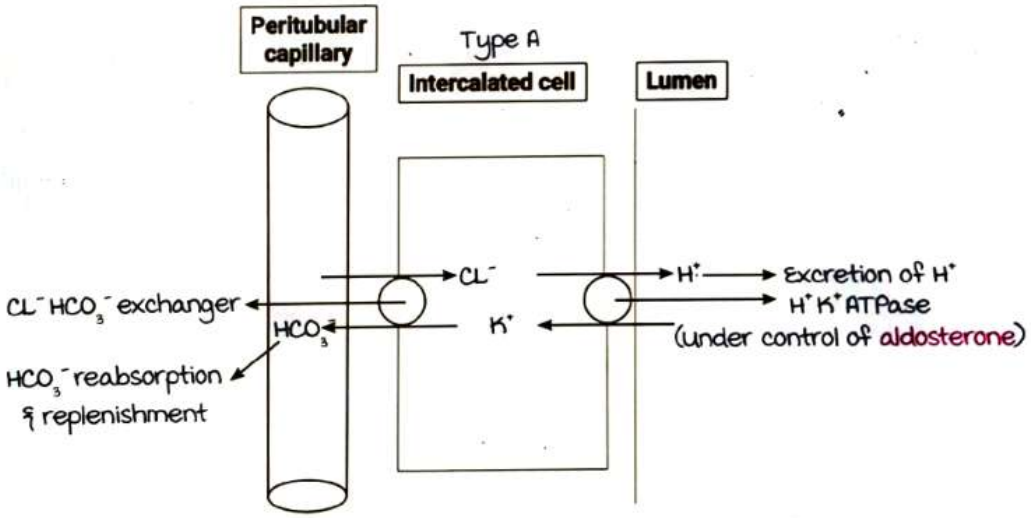
b. with NH_3 as buffer.



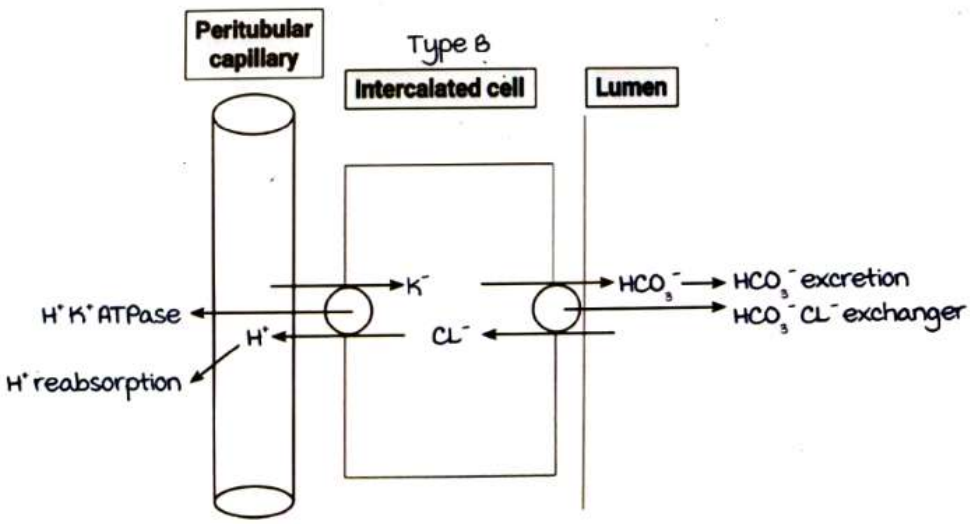
Collecting duct :

Regulated by protein : Hensin.

1. H^+ excretion : In acidosis by intercalated cell A (Acid excretion)



2. HCO_3^- excretion : In alkalosis by intercalated cell B (Base excretion)



Feedback

Www.Medicalstudyzone.com

	Normal values
pH	7.35 – 7.45
pCO_2	35 – 45 mm Hg
HCO_3^-	22 – 26 mEq/L (Avg = 24)

	Causes		Compensation by	
	↑ H^+	↓ HCO_3^-	Lungs	Kidneys
Metabolic acidosis (↓ pH)	• Lactic acidosis. • Diabetic ketoacidosis. • Salicylic acid poisoning.	• Diarrhea (Intestinal content are alkaline).	Hyperventilation → CO_2 washout → ↓ pCO_2 & ↓ H^+	H^+ exchanger : H^+ excretion by intercalated cell A in CD Can lead to hyperkalemia.
	↓ H^+	↑ HCO_3^-	Lungs	Kidneys
Metabolic alkalosis (↑ pH)	• Vomiting (Loss of acidic stomach contents). • Hyperaldosteronism (↑H^+ excretion). • Diuretics → hypovolemia + RAAS → ↑Aldosteronism → H+ excretion → Volume contraction alkalosis.	Sodium bicarbonate ingestion.	↓ respiratory rate and depth (Hypoventilation). → CO_2 build up ↑ pCO_2	Base excretion by type B intercalated cells.

	↑pCO_2	Kidneys
Respiratory acidosis (↓ pH)	• Respiratory depression : Opioids, barbiturates intake. • Respiratory muscle paralysis : Guillain-Barre syndrome, Polio. • Airway obstruction : COPD.	↑ HCO_3^- by : • H+ excretion. • Reabsorption and generation of HCO_3^-.
	↓pCO_2	Kidneys
Respiratory alkalosis (↑ pH)	Hyperventilation due to : • Hysteria. • High altitude.	HCO_3^- excretion by type B I cells → ↓ HCO_3^- levels

Note :

H^+ and Ca^{2+} competitively bind to albumin → ↓H^+ in respiratory alkalosis → ↑ Ca^{2+} binding to albumin → ↓ free form of Ca^{2+} (↓ actions) → Hypocalcemic tetany

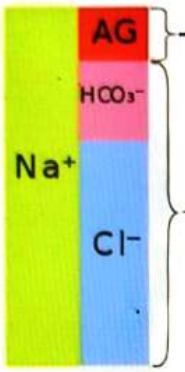
Feedback

Www.Medicalstudyzone.com

----- Active space -----

Anion Gap :

Laws of electroneutrality.



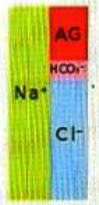
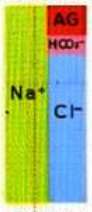
Anion gap : Contributes to electroneutrality

- Difference between anions & cations arising d/t anions not routinely measured.
  - Sulfate.
  - Protein anions.
  - lactate.
  - Salicylate.
- measured by :

$AG = Na^+ - (Cl^- + HCO3^-)$

Normally = 144 − (108 + 24)

= 12 mEq/L (Range : 8–16 mEq/L)

	High anion gap metabolic acidosis (HAGMA)	Normal anion gap metabolic acidosis (NAGMA)
Cause	↑ in unmeasured anions  HAGMA	↑ in measured anions ↑ in Cl^- → Hyperchloremic metabolic acidosis  NAGMA
Examples	• Diabetic Ketoacidosis. • Lactic acidosis. • Salicylate poisoning.	• Diarrhea. • Renal tubular acidosis. • Hyperaldosteronism.

Feedback